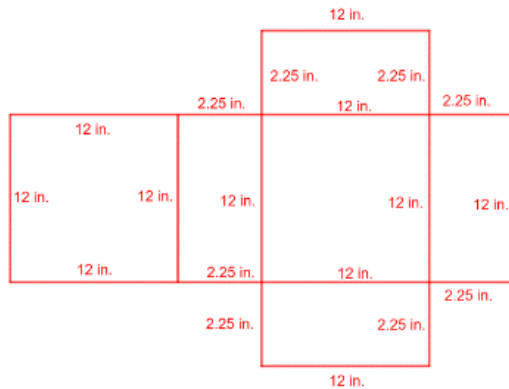


Grade 6 Accelerated
Unit 8
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

A pizza box measures 12 inches (in.) long, 12 in. wide, and 2.25 in. tall.

1. Sketch the net of the pizza box.



2. Label the dimensions of each face on the net.

3. Find the area of each face. Include units.

1. $12 \times 12 = 144 \text{ in.}^2$
2. $12 \times 12 = 144 \text{ in.}^2$
3. $12 \times 2.25 = 27 \text{ in.}^2$

4. $12 \times 2.25 = 27 \text{ in.}^2$
5. $12 \times 2.25 = 27 \text{ in.}^2$
6. $12 \times 2.25 = 27 \text{ in.}^2$

4. Add the area of the faces to find the surface area of the pizza box. Include units.

$$144 + 144 + 27 + 27 + 27 + 27 = 396 \text{ in.}^2$$

5. Geri is wrapping a small gift box for a friend's birthday. The box is 2.75 inches (in.) long, 1.75 in. wide, and 1 in. high. What is the surface area of the box that Geri will need to buy wrapping paper for? Include units.

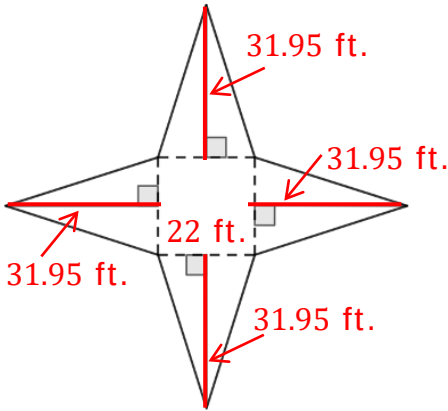
1. $2.75 \times 1.75 = 4.8125 \text{ in.}^2$
2. $2.75 \times 1.75 = 4.8125 \text{ in.}^2$
3. $2.75 \times 1 = 2.75 \text{ in.}^2$
4. $2.75 \times 1 = 2.75 \text{ in.}^2$
5. $1.75 \times 1 = 1.75 \text{ in.}^2$
6. $1.75 \times 1 = 1.75 \text{ in.}^2$

$$4.8125 + 4.8125 + 2.75 + 2.75 + 1.75 + 1.75 = 18.625 \text{ in.}^2$$

In the area that is now known as the Sudan, there is a collection of pyramids known as the Nubian Pyramids. The largest pyramid has a square base of 22 feet (ft) and stands 30 ft high. The slant height is 31.95 ft. The net of the pyramid is shown.

6. Label the dimensions of the net. Include units.
7. Complete the table to determine the area of each face of the pyramid.

Side	Formula	Area
Rectangular Base	$A = l \cdot w$	$A = 22 \times 22 = 484 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
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8. Determine the surface area of the pyramid. Include units.
 $484 + 351.45 + 351.45 + 351.45 + 351.45 = 1,889.8 \text{ ft.}^2$
9. Calculate the surface area of a rectangular pyramid that has a square base with a side length of 11.5 inches (in.), which is the base of the triangle with a slant height of 15.13 in.
 $11.5 \times 11.5 = 132.25 \text{ in.}^2$
 $132.25 + 4(86.9975) = 480.24 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
10. Calculate the surface area of a rectangular pyramid that has a rectangular base with a side length of 4 feet (ft), which is the base of a triangle with a slant height of 3.9 ft, and a side length of 5 ft, which is the base of a triangle with a slant height of 4.1 ft.
 $4 \times 5 = 20 \text{ ft.}^2$
 $20 + 7.8 + 7.8 + 10.25 + 10.25 = 56.1 \text{ ft.}^2$
 $\frac{1}{2} \times 4 \times 3.9 = 7.8 \text{ ft.}^2$
 $\frac{1}{2} \times 4 \times 3.9 = 7.8 \text{ ft.}^2$
 $\frac{1}{2} \times 5 \times 4.1 = 10.25 \text{ ft.}^2$
 $\frac{1}{2} \times 5 \times 4.1 = 10.25 \text{ ft.}^2$